

ELIO VINCIGUERRA

Catania, Italy
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EDUCATION

Ph.D. in Systems Engineering, Energy, Computer Science, and Telecommunications

University of Catania, Italy

Aug. 2023 - Jul. 2026

Master's Degree in Computer Engineering

University of Catania, Italy

Oct. 2021 - Jul. 2023

Thesis: *Analysis and Optimization of Indoor Localization Algorithms for HPC Architectures*

Final grade: 108/110

Bachelor's Degree in Computer Engineering

University of Catania, Italy

Oct. 2018 - Nov. 2021

Thesis: *Evaluation of the Energy-Latency Trade-off in Deep Neural Network Accelerators*

Final grade: 95/110

Diploma in Computer Science and Telecommunications

I.I.S.I.T. G. Marconi, Italy

Sept. 2013 - Jul. 2018

Final grade: 94/100

EXPERIENCE

KTH Royal Institute of Technology

Visiting PhD Student

*Apr. 2026 - May 2026
Stockholm, Sweden*

- Use of the NoCDAS framework and integration of Transformers models and computational NoC paradigm within it.

KTH Royal Institute of Technology

Visiting PhD Student

*Sept. 2025 - Jan. 2026
Stockholm, Sweden*

- Use of the CGRA SiLago framework and integration of Deep Reinforcement Learning algorithms within it.

University of Catania

Nov. 2023 - Jul. 2026

- Computer Architecture Course Teaching Assistance.
- Fundamentals of Computer Science Course Teaching Assistance.
- IoT Systems and Technologies Course Teaching Assistance

Flazio S.r.l.

Intern

*Nov. 2017 - Dec. 2017
Catania, Italy*

- Web Design, website development using the Flazio platform.
- Social media integration.

BaxEnergy S.r.l

Intern

*Jul. 2017 - Aug. 2017
Catania, Italy*

- Quality control of company software programs.

LANGUAGE PROFICIENCY

Italian	Native
English	Intermediate (B2)
Spanish	Intermediate (B1)
Chinese	Basic (A1)

TECHNICAL STRENGTHS

Programming Languages	Python, C, C++, Java, Go, C#
Web Technologies	HTML, CSS, PHP
Databases	SQL (MySQL, PostgreSQL, Microsoft SQL)
Tools	Git, VS Code, IntelliJ, Eclipse

PUBLICATIONS

1. E. Vinciguerra, G. Siracusano, G. Ascia and M. Palesi, "SAFER: Self-Attentive Feature Extraction and Reduction," in *IEEE Access*, vol. 14, pp. 85695-85708, 2026, doi: 10.1109/ACCESS.2026.3700810.
2. E. Vinciguerra, E. Russo, G. Ascia, and M. Palesi, "CHAOS: Controlled hardware fault injector system for gem5," 2026. arXiv: 2602.02119 [cs.AR]. [Online]. Available: <https://arxiv.org/abs/2602.02119>.
3. E. Vinciguerra, M. Palesi, and G. Ascia, "Improving Machine Learning Anomaly Detection for Safety-Critical RISC-V Automotive Systems," in *Proceedings of the CPS Summer School PhD Workshop 2025*, C. Pilato et al., Eds., vol. 4106. Aachen, Germany: CEUR-WS.org, 2025. [Online]. Available: <https://ceur-ws.org/Vol-4106/short1.pdf>.
4. E. Russo, E. Vinciguerra, M. Palesi, D. Patti, G. Ascia and V. Catania, "TeleSABRE: Heuristic Layout Synthesis in Multi-Core Quantum Systems with Teleport Interconnect," 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), Albuquerque, NM, USA, 2025, pp. 749-758, doi: 10.1109/QCE65121.2025.00086.
5. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia, "An Anomaly Detection Model for RISC-V in Automotive Applications: A Domain-Specific Accelerator Perspective," 2025 33rd Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP), Turin, Italy, 2025, pp. 317-320, doi: 10.1109/PDP66500.2025.00051.
6. E. Vinciguerra, E. Russo, M. Palesi and G. Ascia, "Data-Driven Simulation Based Fault Detection in Automotive RISC-V Applications," 2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc), Kuala Lumpur, Malaysia, 2024, pp. 509-516, doi: 10.1109/MCSoc64144.2024.00089.
7. E. Russo, E. Vinciguerra, G. Ascia and D. Panno, "DNN Hardware Accelerator Selection for Feasible Deployment of MARL-Based MAC Protocols in Industrial IoT Networks," 2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI), Milano, Italy, 2024, pp. 443-448, doi: 10.1109/RTSI61910.2024.10761172.
8. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia and H. Rafique, "Improving LSTM-based Indoor Positioning via Simulation-Augmented Geomagnetic Field Dataset," 2024 IEEE International Conference on Mobility, Operations, Services and Technologies (MOST), Dallas, TX, USA, 2024, pp. 251-259, doi: 10.1109/MOST60774.2024.00034.